# ROHIT LOHARKAR *AI Engineer | Software Engineer*

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## SUMMARY

Data Science M.Tech candidate and AI/ML engineer with hands-on experience building LLM-powered applications, RAG workflows, machine learning pipelines, and cloud/infrastructure automation. Strong in Python, GenAI, ML/DL, SQL, Docker, and Linux, with practical exposure to full-stack AI applications and production log analytics.

## TECHNICAL SKILLS

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| **Primary Language** | **Python**, SQL, C++, TypeScript |
| **Agentic AI** | LLMs, RAG, LangChain, LangGraph, CrewAI, Prompt Engineering, ChromaDB |
| **ML / Deep Learning** | Machine Learning, Deep Learning, LSTM, CNNs, MLflow |
| **Cloud / DevOps** | Linux, Docker, Kubernetes, Jenkins, GCP, Azure AI |
| **Databases / API** | MongoDB, DuckDB, SQL, PowerBI, React, FastAPI |

**Certifications:** KodeKloud Linux, Docker, Jenkins, Kubernetes, Python | **Profiles:** Google Cloud Profile | Leetcode Profile

## EXPERIENCE

**Vanderlande Logistics** | **AIML Engineer (Internship)** | *Pune, MH* *Jun 2025 – Jun 2026*

- Built an AI-based RCA system to analyze debug logs, identify probable root causes, and reduce software developers' manual investigation effort by 50%.
- Developed the log intelligence workflow using Python, LangGraph, ChromaDB, Redis, and Azure Blob Storage for scalable ingestion, retrieval, and contextual analysis of industrial logs.
- Contributed to an AI-based test case generation engine that converts functional context and log insights into structured test scenarios for faster validation.
- Built an MCP-based test automation framework using Playwright, OPC UA MCP, and custom MCP servers to automate application and industrial protocol testing workflows.

**ESDS Software Solutions** | **IMS Automation Trainee** | *Nashik, MH* *Feb 2024 – July 2024*

- Assessed Linux server vulnerabilities, prioritized remediation tasks, and supported system hardening activities including service validation, configuration checks, and access control improvements to strengthen reliability and security posture.
- Automated vulnerability checks and operational workflows using Python scripts and Docker, reducing manual effort by 40% while improving consistency in patch validation, testing, and infrastructure support.

## EDUCATION

**Masters of Technology (M.Tech) - Data Science | CGPA: 8.60/10** *Pune, MH*
COEP Technological University *Jul 2024 – Jul 2026*

**Bachelors of Engineering (B.E) - Computer Engineering | CGPA: 8.16/10** *Nashik, MH*
Sandip Institute of Technology and Research Center *Jul 2019 – Jun 2023*
***Honours** in Artificial Intelligence and Machine Learning (AIML)*

## PROJECTS

**EVision - Intelligent EV Eligibility Analytics Platform** | *Python, XGBoost, Scikit-Learn, Pandas, LangChain*

- Built an end-to-end ML pipeline on 150k+ EV registration records to predict Clean Alternative Fuel Vehicle (CAFV) eligibility.
- Used geospatial data, manufacturer trends, electric range, and vehicle attributes to improve prediction quality and model reliability. Added SHAP explainability and reusable inference modules to make predictions easier to interpret and extend.
- Achieved 99.41% accuracy using feature engineering, cross-validation, and Bayesian hyperparameter tuning.

**CipherGuard-AI – FinSpark Hackathon** | *Python, LangGraph, XGBoost, NetworkX, React, TypeScript, FastAPI*

- Built an AI cybersecurity platform correlating network telemetry, authentication logs, and financial transactions to identify sophisticated threats proactively.
- Orchestrated multi-agent investigations with LangGraph and LangChain, combining NetworkX knowledge graphs, XGBoost, Isolation Forest, and SHAP for explainable threat detection.
- Developed a complete React and TypeScript web application with a FastAPI backend, integrating PostgreSQL, FAISS, Pandas, NumPy, Plotly, Docker, and Git for real-time security monitoring and investigation.

## PUBLICATION

**Population-Stratified Variant Pathogenicity Prediction via Ancestry-Conditioned Ensemble Learning and Explainable Clinical Reporting** *Jun 2026*
*ICONDEEPCOM - 2026*

- Developed an ancestry-aware stacking ensemble using XGBoost, LightGBM, and CatBoost to classify genomic variants from a 71-feature dataset, incorporating population-level ancestry signals and patient parameters such as age and sex; generated SHAP-based clinical reports to explain variant pathogenicity predictions and support interpretable clinical decision-making.